Who Carries the Markups? Scalability, Sorting, and the Welfare Cost of Market Power

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Abstract

Standard oligopoly models let firms differ in productivity but assign every firm the same returns to scale. In this paper, I introduce firm-level heterogeneity in returns to scale, disciplined by gross-output production function estimates for Spanish firms in Orbis, into an Atkeson-Burstein-style Cournot oligopoly and study how the sorting between productivity and scalability shapes the welfare cost of market power. The sorting is calibrated to how returns to scale vary with firm size in the data. In the calibrated economy, removing markups raises consumption by 32.3%, more than twice the gain in an economy with common returns to scale, as productivity and scalability act as complements. The bulk of the gap comes from the scale of production rather than from the misallocation of a given stock of inputs, since wedges hold back output most at the firms with the highest returns to scale. Holding both marginal distributions fixed, the gain is S-shaped in the correlation between the two margins, rising from 9.0% under perfect negative pairing to 50.3% under random reassignment. The welfare cost of market power therefore depends on which firms carry the markups.

Keywords: Markups, Returns to scale, Scalability, Sorting, Market power, Misallocation, Oligopoly

1. Introduction

Firm heterogeneity plays a key role in the current literature addressing the rise in market power, in particular in explaining the documented rise in markups over recent decades. Seminal models of firm heterogeneity, such as Melitz (2003), Atkeson & Burstein (2008) and Hsieh & Klenow (2009), make firm size depend on productivity alone, and in Atkeson & Burstein (2008) more productive firms also capture larger market shares and therefore charge higher markups. Heterogeneous markups then distort both the allocation of demand across firms and the overall level of production (Baqaee & Farhi, 2020; De Loecker et al., 2020; Edmond et al., 2023). Such models normally assume homogeneous *returns to scale* (*RTS*) across firms, letting heterogeneity in *total factor productivity* (*TFP*) explain most of the origin of firm markups. Recent works such as Lashkari et al. (2024), Argente et al. (2025), Demirer (2025) and Hubmer et al. (2025), however, have shown that firms differ in how easily they scale, and that this heterogeneity plays an important role in explaining key facts of firm dynamics previously associated with TFP differences. In particular, Hubmer et al. (2025) find in their quantitative implementation that heterogeneity in RTS more than doubles the efficiency losses from financial frictions, compared with an economy where firms differ in TFP alone.

In this paper, I introduce firm-level heterogeneity in returns-to-scale parameters to a standard Atkeson & Burstein (2008)-style Cournot oligopoly framework, and explore how important this joint heterogeneity of productivity and scalability and its alignment are for the welfare impact of market power. With heterogeneous RTS, productivity and scalability jointly determine equilibrium size, and, under oligopoly, size determines markups. I argue that the sorting between fundamental productivity, which determines a firm's inverse marginal cost at a common reference output, and scalability, and not only their absolute value, plays a major role in shaping the aggregate welfare costs of market power. The mechanism is that a given markup wedge is more distortionary when it is attached to a firm that can absorb large quantities with only a limited increase in cost.

To achieve this, I introduce a quantitative framework designed to incorporate both variation in productivity and RTS within a pre-established and well-known oligopoly environment. The setting is an Atkeson-Burstein-style sector oligopoly model with heterogeneous market exposure and finite-firm competition inside product markets. Producers are draws from a common continuous joint distribution $F(\alpha, \nu)$ where the RTS margin α is externally disciplined, while the productivity margin ν has a Pareto marginal with tail index ξ following Edmond et al. (2023). Returns to scale are disciplined by firm-level gross-output production function estimates based on the approach of Gandhi et al. (2020) (henceforth GNR) in the spirit of Hubmer et al. (2025). Meanwhile, firm productivity is summarized by its inverse marginal cost at a common anchor output, calibrated inside the model. Productivity and scalability are linked by a rank copula whose strength $\bar{\rho}$ is also calibrated, and the number of firms in each sector is drawn from a Poisson distribution with mean N. The within-market elasticity γ mainly disciplines the aggregate markup, which I measure following the production-function approach of De Loecker et al. (2020), as the product of the estimated variable-input elasticity and the firm's ratio of sales to materials and staff costs, aggregated across firms with cost weights that include the user cost of capital, yielding a target of 1.1815. I use cost rather than sales weights since, as Edmond et al. (2023) show, the sales-weighted markup equals the cost-weighted one plus a term that captures the variance of markups, and would therefore overstate the aggregate wedge.

I use firm-level data for Spanish firms in Orbis from 2015 to 2019, with monetary values expressed in 2019 euros, which leaves me with 1,391,061 firm-years for the estimation of returns to scale. For each NAICS2 cell, I estimate a gross-output production function with the GNR method, whose materials, labor and capital output elasticities vary across firm-years with their input mix, and sum the three into a firm-year RTS estimate, whose mean over the window

forms the scalability distribution F^{α}[1]. I do not target concentration measures, since measured concentration in Orbis depends on the representativeness of the firm sample (Kalemli-Özcan et al., 2024) and on the market definition, and instead discipline the upper tail of firm size through two tail indices of firm sales.

In line with Hubmer et al. (2025), I find that returns to scale in the Orbis data are U-shaped in firm size, both within NAICS2 industry-years and in the pooled sample, with the top 5% of firms by sales reaching an α about 0.074 above the industry mean. As in their estimates, the steep rise at the top is driven by the output elasticity of intermediate inputs. Differently from the authors, however, the left arm of the U is more pronounced, since the labor elasticity falls with size.

The calibrated parameters are pinned down jointly to match the aggregate markup, the two tail indices of firm sales and the gap in returns to scale between the largest firms and the bottom half of the sales distribution, which identifies the sorting parameter. As in Hubmer et al. (2025), I calibrate the dependence between productivity and scalability internally, since their joint distribution cannot be read off the production-function estimates, and like the authors, I find a negative correlation between the two, such that some highly scalable firms remain small while others reach the top of the sales distribution, in line with the U-shaped profile above.

Removing every markup raises steady-state consumption by $\lambda = 32.3\%$ in the calibrated economy[2], which I refer to as the welfare gap between the market and the planner allocation. To understand which dimension of heterogeneity drives this gap, I compare this baseline (α, ν) -economy with three counterfactual economies, each recalibrated to the targets it can reach, such that the comparison holds the observed moments fixed rather than the parameters. The ν-economy holds RTS common at the harmonic mean of F^{α}, as in the standard approach to measuring the cost of markups (Edmond et al., 2023), and matches the same markup level and nearly matches the same sales tails as the baseline. However, its welfare gap is only 14.3%, less than half of the calibrated one. The α-economy and the homogeneous economy, which both hold productivity common, reach 13.0% and 10.8%, such that each margin on its own adds little to the cost of a common markup, with productivity dispersion contributing 3.48 percentage points and scalability dispersion 2.19. I find that the joint interaction of the two, by contrast, accounts for 15.6 percentage points, or 48.7% of the calibrated gap. That is, productivity and scalability act as complements, and almost half of the welfare cost of markups arises only when firms differ along both dimensions at once.

To understand where this cost comes from, I split the welfare gap in two separate ways using intermediate planner allocations. The first separates the dispersion of markups across firms from their common level (Baqaee & Farhi, 2020; Edmond et al., 2023), and attributes 43% of the gap to dispersion. The second asks whether the planner gains by reallocating the inputs that the market already uses or by using more of them, in the spirit of Hsieh & Klenow (2009) and Restuccia & Rogerson (2008), and finds that reallocation alone would raise consumption by only 5.5%, while capital accounts for 63% of the gap. Therefore, markups in this economy lower welfare mostly by holding back the expansion of scalable firms, which corroborates the idea that the two margins act as complements. Productivity generates the fat tail of sales, and with it the large shares and high Cournot markups, while scalability determines how much output a given markup withholds, since the elasticity of a firm's output to its own markup rises with its returns to scale (Ruzic & Ho, 2023), going from $\gamma = 7.88$ under constant returns to about 28 at $\alpha = 1.1$.

[1]See Section 3.2 for more information. Since cost minimization requires returns to scale not to exceed the markup, scalability draws come from F^{α} truncated at $0.99\gamma/(\gamma - 1)$, just below the CES floor.

[2]This value comes from the sorting experiments of Section 4.3, which are solved on $M = 10000$ markets. The comparison across economies in Table 2 uses $M = 5000$, where the calibrated gain is 32.04%, and the percentage-point contributions and shares reported for that comparison and its decomposition in Figure 2 refer to this value.

To measure how much the pairing matters for this complementarity, I hold the calibrated parameters and both marginal distributions fixed and vary only the strength of the rank copula, together with within-sector permutations that implement perfect negative and positive pairing and random reassignment. I find that the pairing moves the welfare gap by far more than any dimension of heterogeneity, with λ rising from 9.0% under perfect negative pairing to 50.3% under random reassignment and 152.3% under perfect positive pairing. The response is S-shaped in the correlation between scalability and productivity and steepest around the calibrated economy, such that λ doubles as the correlation moves from -0.35 to the calibrated -0.14, which makes the RTS gap that identifies sorting a first-order determinant of the cost of markups. Concluding that positive sorting lowers welfare would, however, be incorrect, since welfare rises with sorting in both the market and the planner allocation, with the gap widening because the planner rises faster.

Using the same two welfare gap decompositions as before, I find that sorting raises the cost of markups through two channels. The first is market power, since aligning scalability with productivity lets the most productive firms expand further, raising both the aggregate markup and the spread of markups. The second is the reallocation of inputs towards scalable firms, whose output responds most to a given markup. Consequently, the dispersion component grows from zero under strongly negative sorting to about half of the gap at the top, and misallocation to about a quarter, while capital remains the largest component throughout.

Related Literature. This paper contributes to three strands of literature. The first quantifies the aggregate cost of markups when firms are heterogeneous, where Edmond et al. (2015), Peters (2020), Edmond et al. (2023) and De Loecker et al. (2021) measure it with a common returns to scale across firms. I add heterogeneity in returns to scale disciplined by production-function estimates, and show that the same markups are far more costly when they sit on scalable firms. The second strand documents that firms differ in their returns to scale. Hubmer et al. (2025) and Demirer (2025) estimate production functions whose output elasticities, and thus returns to scale, vary across firms, with Hubmer et al. (2025) showing that returns to scale rise with size at the top of the distribution, while Kariel & Savagar (2024) and Baqaee et al. (2024) study the aggregate implications of scale economies. A related literature traces scalability to information technology (Lashkari et al., 2024), scalable expertise (Argente et al., 2025) and intangible inputs (De Ridder, 2024). I contribute to this strand by embedding returns to scale estimated on Spanish firms in an oligopoly with endogenous markups, which shows that returns to scale determine how much output a given markup withholds, and therefore the welfare cost of market power. Finally, Restuccia & Rogerson (2008), Hsieh & Klenow (2009) and Baqaee & Farhi (2020) show that wedges matter for aggregate outcomes through their covariance with producer characteristics, while David & Venkateswaran (2019), Ruzic & Ho (2023) and Zhang & Xia (2022) show that heterogeneity in technologies and markups changes how misallocation should be measured. In this paper, the sorting between productivity and scalability endogenously places the largest markups on the firms that can expand most easily, which accounts for a substantial share of the welfare gap.

2. Theory

2.1. Households

Household problem. Time is discrete. There is a unit mass of identical households. The representative household consumes the final good C_t, supplies labor L_t, owns a diversified portfolio of firms, and rents capital services to producers at the competitive rental rate R_t. Firms also purchase the aggregate composite as materials. The household accumulates capital according to

$$K_{t+1} = (1 - \delta_K)K_t + I_t,$$

The intertemporal first-order condition equates the marginal rate of substitution to the net return on capital, $U_{C(C_t,L_t)} = \beta E_t[U_C(C_{t+1}, L_{t+1})(R_{t+1} + 1 - \delta_K)]$, so in any steady state the rental rate is pinned by preferences and depreciation alone:

$$R = \frac{1}{\beta} - (1 - \delta_K). \tag{1}$$

Equation (1) holds in the market allocation and in the planner allocation alike: neither the markup wedge nor the allocation of capital across firms moves the household's required return. Capital is therefore supplied perfectly elastically at R, and the aggregate stock K is a residual of firm demand rather than a constraint. The household maximizes lifetime utility

$$E_0 \sum_{t=0}^{\infty} \beta^t U(C_t, L_t),$$

with period utility

$$U(C_t, L_t) = \ln C_t - \chi \frac{L_t^{1+\frac{1}{\varphi}}}{1 + \frac{1}{\varphi}}, \quad \chi > 0, \varphi \geq 0.$$

Aggregate composite. The aggregate composite[3] is produced from a continuum of product markets $j \in [0, 1]$ with exposure weights ω_j:

$$Q_t = \left[\int_0^1 \omega_j^{\frac{1}{\eta}} Y_{j,t}^{\frac{\eta-1}{\eta}} dj \right]^{\frac{\eta}{\eta-1}}, \quad \eta > 1, \tag{2}$$

with associated price index

$$P_t = \left[\int_0^1 \omega_j P_{j,t}^{1-\eta} dj \right]^{\frac{1}{1-\eta}}. \tag{3}$$

Gross composite output is split between household consumption, investment and intermediate use, such that

$$C_t = Q_t - M_t - I_t, \quad M_t = \int_0^1 \sum_{i \in A_{j,t}} m_{ji,t} dj, \tag{4}$$

which in a steady state, where $I = \delta_K K$, reads $C = Q - M - \delta_K K$. The exposure weight ω_j summarizes how much aggregate expenditure is exposed to product market j and is normalized so that $\int_0^1 \omega_j dj = 1$. In the current baseline the simulated sectors are pooled with normalized exposure weights so they all have the same weight. Materials are purchases of the same composite at price P_t. The expenditure allocated to market j is $X_{j,t} \equiv P_{j,t} Y_{j,t}$, and the corresponding demand schedule loads on total composite use:

$$Y_{j,t} = \omega_j \left(\frac{P_{j,t}}{P_t} \right)^{-\eta} Q_t.$$

[3]**Notation.** Sector, market, and firm indices are written as subscripts: Y_j, P_j, ω_j, A_j, n_j for sector-level objects, and s_{ji}, μ_{ji}, y_{ji}, p_{ji} for firm i in sector j. This matches the numerical implementation, where sectors are a finite simulated sample $j = 1, ..., J$ and aggregates are sums $\sum_{j=1}^{J}$. In the underlying theory, however, sectors are a continuum $j \in [0, 1]$, so these objects are properly fields over the unit interval ($Y(j)$, $P(j)$, $\omega(j)$, $s(j, i)$, and so on), with population aggregates given by integrals $\int_0^1 \cdot \, dj$ rather than sums. The subscript form is used purely for readability; it carries no discreteness assumption at the theory level.

2.2. Demand and Market Shares

Nested demand structure. There is a continuum of finite sectors indexed by $j \in [0,1]$. Each sector is one product market. Let $X_{j,t}$ denote composite expenditure assigned to sector j. Within sector j a finite number $n_{j,t}$ of firms compete. Firm $i \in \{1, ..., n_{j,t}\}$ supplies quantity $y_{ji,t}$ and charges price $p_{ji,t}$. Conditional on sector expenditure, the within-sector composite good is CES. Throughout the baseline I write the aggregator and its price index with an explicit $1/n_{j,t}$ weight that removes mechanical love of variety[4], where $n_{j,t} \equiv |A_{j,t}|$ is the realized number of allocated producing firms (the set $A_{j,t}$ is defined in Section 2.4 below):

$$Y_{j,t} = \left[\sum_{i \in A_{j,t}} \frac{1}{n_{j,t}} y_{ji,t}^{\frac{\gamma-1}{\gamma}} \right]^{\frac{\gamma}{\gamma-1}}, \quad \gamma > \eta, \tag{5}$$

with price index

$$P_{j,t} = \left[\left(\frac{1}{n_{j,t}} \right)^{\gamma} \sum_{i \in A_{j,t}} p_{ji,t}^{1-\gamma} \right]^{\frac{1}{1-\gamma}} = n_{j,t}^{\frac{\gamma}{\gamma-1}} \left[\sum_{i \in A_{j,t}} p_{ji,t}^{1-\gamma} \right]^{\frac{1}{1-\gamma}}.$$

The within-market elasticity of substitution γ is common across markets: it governs how aggressively firms inside a market substitute against one another. The across-market elasticity η is a single global object governing substitution across product markets.

Shares and markups. The conditional demand for firm i in market j is therefore

$$y_{ji,t} = \left(\frac{1}{n_{j,t}} \right)^{\gamma} \left(\frac{p_{ji,t}}{P_{j,t}} \right)^{-\gamma} Y_{j,t}, \tag{6}$$

and its revenue share inside the market is

$$s_{ji,t} \equiv \frac{p_{ji,t} y_{ji,t}}{\sum_h p_{jh,t} y_{jh,t}} = \frac{p_{ji,t}^{1-\gamma}}{\sum_h p_{jh,t}^{1-\gamma}}. \tag{7}$$

Finite-firm competition implies that firms internalize the effect of their own quantity choices on the market price index. In the nested-CES Cournot environment studied here and by Edmond et al. (2023) and by Atkeson & Burstein (2008), this generates an inverse-markup rule of the form

$$\frac{1}{\mu_{ji,t}} = 1 - \frac{1}{\gamma} - \left(\frac{1}{\eta} - \frac{1}{\gamma} \right) s_{ji,t}. \tag{8}$$

Equation (8) is the key pricing object in the model. Because $\gamma > \eta$, the coefficient on $s_{ji,t}$ is negative, implying that firms with larger market shares face less elastic residual demand and charge higher markups. With a common γ, the within-sector markup floor as $s_{ji,t} \to 0$ is the same across sectors, $\mu_{\min} = \frac{\gamma}{\gamma-1}$; all heterogeneity in realized markups comes from endogenous market shares, which in turn reflect the distribution of firm draws within each sector and the exogenous sector firm count. The detailed Cournot derivation is relegated to the appendix.

[4]Under symmetry, n firms each producing y would deliver an unweighted composite $n^{\frac{\gamma}{\gamma-1}} y$, which rises with the number of firms at fixed per-firm output. The weight removes this term and yields $Y_{j,t} = y$, so that only reallocation and heterogeneity gains remain. Conditional on sector expenditure $X_{j,t}$, it rescales $Y_{j,t}$ and $P_{j,t}$ by reciprocal powers of $n_{j,t}$ and leaves every firm-level quantity, price, share, markup and profit unchanged. Only the within-sector nest carries this normalization, since the exposure weights ω_j are fixed and Equation 2 has no variety margin.

2.3. Firms and Technology

Heterogeneity. Firms differ along two dimensions. First, they differ in scalability, indexed by α_{ji}, which governs gross-output returns to scale and how fast marginal cost rises with output. Second, they differ in fundamental productivity, indexed by ν_{ji}, the inverse-marginal-cost shifter that fixes each firm's marginal cost at a common anchor output $\hat{y}$: a higher ν_{ji} means a lower marginal cost at the anchor. Fundamental productivity ν_{ji} is distinct from the production-function TFP coefficient z_{ji} introduced below. The empirical strategy treats α_{ji} as externally disciplined and treats ν_{ji} as the productivity primitive. Henceforth, productivity refers to ν_{ji}. The dependence between α and ν is introduced only through ranks. Let $\tilde{\alpha}_{ji}$ be the standard-normal score of firm i's scalability rank in sector j, and let $\varepsilon_{ji} \sim N(0,1)$. The productivity ladder is

$$\ell_{ji} = \bar{\rho}\tilde{\alpha}_{ji} + \sqrt{1-\bar{\rho}^2}\varepsilon_{ji}, \tag{9}$$

and the ladder percentile is mapped into a Pareto productivity draw,

$$u_{ji} = \Phi(\ell_{ji}), \quad \nu_{ji} = \underline{\nu}(1-u_{ji})^{-\frac{1}{\xi}}, \quad \xi > 1. \tag{10}$$

The Pareto tail index ξ is calibrated to the tail indices of firm sales, the lower bound $\underline{\nu}$ is a units-fixing scale solved by the baseline anchoring condition (not moment-targeted), and the alignment of ν with α is set through the calibrated rank-copula parameter $\bar{\rho}$, which fixes the strength of the α–ν rank copula. It is identified by the gap in mean returns to scale between the top 5% and the bottom 50% of firms by sales, 0.046, a directly observed cross-sectional moment requiring no productivity proxy and no data-side level anchor (see Section 3.3).

Technology and Anchoring. Let ϕ_v denote the value-added weight. Production in firm i of market j is gross output, combining the value-added composite $v_{ji,t} \equiv k_{ji,t}^{a} l_{ji,t}^{1-a}$ with materials $m_{ji,t}$ in a constant-returns Cobb-Douglas input bundle $x_{ji,t} \equiv v_{ji,t}^{\phi_v} m_{ji,t}^{1-\phi_v}$:

$$y_{ji,t} = z_{ji} x_{ji,t}^{\alpha_{ji}}, \quad x_{ji,t} = v_{ji,t}^{\phi_v} m_{ji,t}^{1-\phi_v}, \quad v_{ji,t} = k_{ji,t}^{a} l_{ji,t}^{1-a}, \quad \phi_v \in (0,1). \tag{11}$$

The parameter a is the common capital share inside value added. The value-added weight ϕ_v and materials weight $1-\phi_v$ give gross-output elasticities $\frac{\partial \ln y}{\partial \ln m} = (1-\phi_v)\alpha_{ji}$, $\frac{\partial \ln y}{\partial \ln k} = \phi_v a \alpha_{ji}$, and $\frac{\partial \ln y}{\partial \ln l} = \phi_v(1-a)\alpha_{ji}$, summing to α_{ji}. The anchored parameterization takes fundamental productivity ν_{ji} as the primitive and treats the TFP coefficient z_{ji} as a derived object, with the two being linked by the relabeling identity

$$z_{ji} = \left(\nu_{ji}/\alpha_{ji}\right)^{\alpha_{ji}} \hat{y}^{1-\alpha_{ji}}, \tag{12}$$

so (11), the input bundle, and the conditional input demands are algebraically unchanged. Only the distributional assumption moves: the common Pareto is now asserted on the dimensionally coherent cost shifter ν_{ji} (marginal cost at the common anchor $\hat{y}$) rather than on z_{ji}, whose units differ across firms when α_{ji} differs.

Cost minimization. Cost minimization in the inner value-added problem implies the common unit cost index

$$\Omega(W_t, R_t) = \left(\frac{R_t}{a}\right)^{a} \left(\frac{W_t}{1-a}\right)^{1-a}. \tag{13}$$

With materials purchased at price P_t, the gross-output unit cost is

$$\Omega_t^g \equiv \Omega^g(W_t, R_t, P_t) = \left(\frac{\Omega(W_t, R_t)}{\phi_v}\right)^{\phi_v} \left(\frac{P_t}{1-\phi_v}\right)^{1-\phi_v}. \tag{14}$$

Given the production technology in (11), expressed through the productivity primitive ν_{ji} at the anchor $\hat{y}$, variable cost can be written in anchored form as

$$TC_{ji,t}(y) = \alpha_{ji}\frac{\Omega_t^g}{\nu_{ji}}\hat{y}\left(\frac{y}{\hat{y}}\right)^{\frac{1}{\alpha_{ji}}}, \tag{15}$$

which delivers marginal cost

$$MC_{ji,t}(y) = \frac{\Omega_t^g}{\nu_{ji}}\left(\frac{y}{\hat{y}}\right)^{\frac{1}{\alpha_{ji}}-1}. \tag{16}$$

At the anchor $y = \hat{y}$, marginal cost is $MC_{ji,t}(\hat{y}) = \Omega_t^g/\nu_{ji}$ regardless of α_{ji}: productivity alone ranks firms at the anchor, and α_{ji} only tilts how marginal cost moves as output deviates from it, so the level channel and the curvature channel are separated by construction.[5] Cost minimization also gives the conditional input demands

$$P_t m_{ji,t} = (1-\phi_v)TC_{ji,t}, \quad W_t l_{ji,t} + R_t k_{ji,t} = \phi_v TC_{ji,t}, \quad \frac{k_{ji,t}}{l_{ji,t}} = \frac{a}{1-a}\frac{W_t}{R_t}. \tag{17}$$

Equation (16) makes the role of scalability transparent. The scale term $\left(\frac{y}{\hat{y}}\right)^{\frac{1}{\alpha_{ji}}-1}$ shows that a larger α_{ji} flattens marginal cost with respect to output around the anchor and allows the firm to expand more aggressively before costs rise sharply.

Pricing and profits. Conditional on demand, the equilibrium pricing condition is

$$p_{ji,t} = \mu_{ji,t}MC_{ji,t}(y_{ji,t}). \tag{18}$$

Profits can then be written as

$$d_{ji,t} = p_{ji,t}y_{ji,t} - TC_{ji,t}(y_{ji,t}) = \left(1 - \frac{\alpha_{ji}}{\mu_{ji,t}}\right)p_{ji,t}y_{ji,t}. \tag{19}$$

Together, (6), (8), and (18) form a fixed-point system inside each market: quantities determine shares, shares determine markups, and markups feed back into prices and quantities through marginal cost.

2.4. Market Structure and Decentralized Equilibrium

Market Structure. Each sector j receives an exogenous number of producing firms,

$$n_{j,t} = \max(1, \text{Poisson}(N)), \tag{20}$$

where N is a calibrated primitive and is interpreted as the mean number of firms per sector[6]. Conditional on this draw, the set $A_{j,t} = \{1, ..., n_{j,t}\}$ contains the firms priced in the sector Cournot problem. All allocated firms produce in the static cross section. There is no per-period fixed production cost and no profitability-based removal of firms.

Equilibrium. Given the aggregate state K_t, the exogenous sector counts, and the distributions of (α_{ji}, ν_{ji}), a decentralized equilibrium is a sequence of household allocations, firm allocations, prices, market shares, and sector firm sets such that (i) households optimize given prices and firm payouts, (ii) firms satisfy demand (6) and markup pricing (18), (iii) each sector firm set $A_{j,t}$ is the exogenous Poisson allocation in (20), (iv) composite feasibility holds as in (4),

[5] The anchored form is an algebraic identity, since writing the original total cost $\Omega_t^g\left(\frac{y}{z_{ji}}\right)^{\frac{1}{\alpha_{ji}}}$ as $\Omega_t^g\left(\frac{\hat{y}}{z_{ji}}\right)^{\frac{1}{\alpha_{ji}}}\left(\frac{y}{\hat{y}}\right)^{\frac{1}{\alpha_{ji}}}$ and substituting Equation 12 gives Equation 15.

[6] The numerical implementation reserves H firm slots per market, with H large enough that the Poisson count rarely reaches it.

with materials clearing through $M_t = \int_0^1 \sum_{i \in A_{j,t}} m_{ji,t} dj$ and the aggregate price level normalized by $P_t \equiv 1$, and (v) labor and capital services clear,

$$L_t = \int_0^1 \sum_{i \in A_{j,t}} l_{ji,t} dj, \tag{21}$$

$$K_t = \int_0^1 \sum_{i \in A_{j,t}} k_{ji,t} dj. \tag{22}$$

Aggregate accounting closes because gross sales equal $P_t Q_t$, variable cost equals factor payments plus materials purchases, and profits are the residual. Hence value added satisfies $P_t(Q_t - M_t) = P_t(C_t + I_t) = W_t L_t + R_t K_t + \Pi_t$, since materials net out as intra-economy purchases and (4) assigns the rest of the composite to consumption and investment.

2.5. Planner's Allocation and Welfare

Planner Problem. The planner faces the same technology, the same exogenous sector firm sets, and the same distributions of scalability and latent productivity. The difference is that the planner does not treat markup pricing as a technological constraint. Instead, the planner allocates labor, capital services, materials, and output so as to maximize household welfare subject only to feasibility, the production technology, and the same allocated firms. Formally, conditional on the producing firm set $A_{j,t}$ in each market, the planner chooses sequences

$$\left\{C_t, Y_{j,t}, y_{ji,t}, l_{ji,t}, k_{ji,t}, m_{ji,t}\right\}_{t \geq 0}$$

to maximize

$$W = E_0 \sum_{t=0}^{\infty} \beta^t U(C_t, L_t),$$

subject to the household aggregator (2), the market bundles (5), the resource constraint (4), the production technology (11), and the aggregate labor and capital constraints (21) and (22). Let λ_t^L and λ_t^K denote the multipliers on the aggregate labor and capital constraints, and let

$$\psi_{ji,t} \equiv \frac{\partial Q_t}{\partial y_{ji,t}}$$

denote the planner's marginal contribution of one additional unit of firm output to the gross composite bundle; since $C_t = Q_t - M_t - I_t$, this is also the marginal contribution to consumption holding materials and investment fixed. The planner's allocation conditions are then

$$U_{C(C_t,L_t)} \psi_{ji,t} \frac{\partial y_{ji,t}}{\partial l_{ji,t}} = \lambda_t^L, \tag{23}$$

$$U_{C(C_t,L_t)} \psi_{ji,t} \frac{\partial y_{ji,t}}{\partial k_{ji,t}} = \lambda_t^K, \tag{24}$$

$$U_{C(C_t,L_t)} \psi_{ji,t} \frac{\partial y_{ji,t}}{\partial m_{ji,t}} = U_{C(C_t,L_t)}. \tag{25}$$

The planner equalizes the social marginal value of labor and capital across product lines, accounting for heterogeneity in scalability and productivity, and uses materials until their marginal product equals the value of the composite diverted from consumption. Because $\psi_{ji,t}$ comes from the same demand system as the decentralized equilibrium, planner and market share a common inverse-demand structure and differ only in the markup wedge in pricing[7].

Regime invariance and the welfare scale. Every allocation considered below is solved under the same two normalizations, $P = 1$ and $L = 1$, and at the same required return (1). Labor is therefore pinned across regimes while capital is free in both the market and the planner allocation: by (27), the aggregate capital stock is whatever firms demand at R. Lifetime welfare W^x of an allocation x is the household objective evaluated at that allocation, and the consumption-equivalent gain from moving from the market (ME) to the planner (PE) allocation is the scalar λ that, applied to market consumption in every period, raises market welfare to W^{PE}. The labor-disutility weight χ is held at its baseline market value in every regime, and since $L = 1$ in every regime, the disutility term is common, such that with log utility over consumption the gain reduces exactly to

$$1 + \lambda = \frac{C^{PE}}{C^{ME}}. \tag{26}$$

Because the planner keeps the producing firm sets A_j of the market[8], λ measures the welfare cost of markup pricing at given producers, which I quantify for the calibrated economy in Section 4 and trace back there to the margins of the allocation that markups distort.

3. Quantification

3.1. Pooled-Market Steady State

Inner-market and Pooled aggregation block. I now characterize the steady state as a nested system. Time subscripts are dropped, the aggregate price index is normalized to $P = 1$, and the distorted market steady state is written under the labor normalization $L = 1$. The resulting equilibrium is then used to pin down the labor-disutility weight χ, after which the market and planner steady states can be compared in common units. Conditional on sector j, the exogenous firm count n_j, and the realized firm draws $\left(\alpha_{ji}, \nu_{ji}\right)_{i \in A_j}$, the steady-state market equilibrium solves jointly for firm quantities, prices, shares, markups, profits, and input demands through (6), (8), (18), (19), and (17). The α draw comes from F^{α} and the productivity draw ν follows (9) and (10). Sector outcomes aggregate through the exposure-weighted CES bundle in (2), with equal ω_j across markets in the quantitative implementation. Consumption is $C = Q - M - \delta_K K$, net of materials and replacement investment.

Capital and factor-market block. Aggregating the conditional input demands (17) over the continuum of markets gives aggregate factor use as a function of aggregate variable cost $TC \equiv \int_0^1 \sum_{i \in A_j} TC_{ji} dj$ alone. Only the fraction ϕ_v of variable cost is a primary-factor payment; the remaining $1 - \phi_v$ buys materials of the composite at $P = 1$:

$$K = \frac{a\phi_v \text{ TC}}{R}, \quad L = \frac{(1-a)\phi_v \text{ TC}}{w}, \quad M = (1-\phi_v) \text{ TC}, \quad C = Q - M - \delta_K K. \tag{27}$$

With R fixed at (1), (27) makes the aggregate capital stock a residual: whatever quantity firms demand at the Euler rate is supplied. The anchored steady-state equilibrium is therefore a three-dimensional fixed point in $(w, X, \underline{\nu})$, where X is composite expenditure per market and $\underline{\nu}$ is the productivity scale. The three residuals are the price normalization $P = 1$, the labor normalization $L = 1$, and the median active-firm output condition $\tilde{y} = \hat{y} \equiv 1$. The rental rate is not among the unknowns.

[7] Appendix A.2 gives $\psi_{ji,t}$ explicitly and maps it through the nested CES structure.

[8] Because the static cross section has exogenous firm counts, the planner does not re-select firms, so a full entry counterfactual is left outside the welfare exercise. The rationale behind this omission is the Edmond et al. (2023) finding that the entry wedge is quantitatively small relative to the markup-level and misallocation channels.

Normalization and steady-state welfare block. At each candidate parameter vector, the distorted market steady state is anchored by the normalizations $P = 1$ and $L = 1$, together with the baseline anchoring condition that the median active-firm output equals the common anchor, $\tilde{y} = \hat{y} \equiv 1$, imposed as one equilibrium condition of the baseline market solve[9]. Conditional on assigned parameters and on the anchored productivity distribution, the market solve pins down the wage-consumption ratio and therefore the labor-disutility weight $\chi = \frac{w}{C}$, using the market labor-leisure condition under $L = 1$ and $C = Q - M - \delta_K K$. Once this baseline calibration step is complete, the planner and market steady states are re-solved in common units while holding the primitive objects fixed. The planner and market allocations therefore share the same $(\beta, \delta_K, a, \phi_o, \hat{y}, \alpha_{ji}, \nu_{ji}, N, \xi, \gamma, \eta, \bar{\rho}, H)$ and differ only through the presence or absence of the aggregate-markup and dispersion distortions. This is what makes the later welfare comparison interpretable as a comparison across allocations rather than across calibrations.

Frozen-anchor protocol. The anchor $\hat{y}$ and all firm draws (α_{ji}, ν_{ji}) are fixed at the baseline and held frozen across the planner solve and every intermediate allocation of Section 4.2: technology is then an ordinary allocation-independent object, so the planner faces literally the same $TC_{ji,t}(y)$ as the market and re-anchoring under the planner is structurally impossible. The recalibrated economies of Section 4.1 keep the same definition of the anchor, the median active-firm output, but solve it on their own draws, such that its value changes with each economy, while the sorting arrangements of Section 4.3 keep the baseline value. This follows the logic of the normalized-CES literature (Klump & La Grandville, 2000), in which technology families are normalized to coincide at a common benchmark before varying curvature parameters. The fixed-$\hat{y}$ construction applies the same identification principle to heterogeneous returns to scale: all technologies agree on marginal cost at the reference output, so differences in α affect only the curvature of marginal cost away from that operating point. I go beyond Klump & La Grandville (2000) by arguing that the benchmark must also be economically sensible, and the median is preferable to the sales-weighted mean in skewed distributions. Active-firm output is heavily right-skewed, so the sales-weighted mean is pulled deep into the tail of that distribution, exceeding the output of roughly 99.6% of active firms in the baseline draw, such that anchoring there would make α_{ji} act, for nearly the entire firm population, as a mechanical penalty for being smaller than a handful of superstar firms. The median sidesteps this by construction, since it splits the active-firm population evenly on either side of $\hat{y}$ regardless of how fat the right tail is, so α_{ji} is disciplined by curvature around a typical operating scale rather than by distance from the largest incumbents.

3.2. Data and Measurement

Sample, Sales, and Markups. I use Spanish Orbis firm-years from 2015–2019, with monetary values expressed in 2019 euros. The sample excludes NAICS2 codes 52/53/62/81 following Hubmer et al. (2025), 92/99 for being out of scope, and 55 for consisting of holding companies and head offices. Manufacturing (31–33), retail (44–45), and transportation (48–49) are grouped into single cells, leaving 14 NAICS2-based cells. The sales and markup sample contains 2,000,342 firm-years from 568,920 firms.

To measure the upper tail of firm size, I average each firm's real sales over the window and divide by the mean of these firm means within its NAICS2 cell. I then pool the normalized sales and estimate the tail index ζ as the negative slope of $\ln\left(r - \frac{1}{2}\right)$ on log normalized sales, using separately the largest 5% and 2% of firms ranked by sales. The resulting indices are 1.1079 and 1.1451, respectively.

Firm-level markups follow the production-function approach of De Loecker et al. (2020). Each firm-year markup is the cell-year estimated variable-input elasticity times the firm's sales divided

[9]This condition pins the productivity scale $\underline{\nu}$ (the Pareto lower bound) as a by-product of its level normalization.

by its materials and staff costs. I weight these markups by materials and staff costs plus capital user cost within each year, then average the five annual values. The resulting aggregate markup target is 1.1815.

Scalability and Firm Size. The distribution F^{α} is not calibrated. It is built once in the empirical pipeline and the solver samples draws from its support. The firm-level object is the gross-output returns-to-scale estimate,

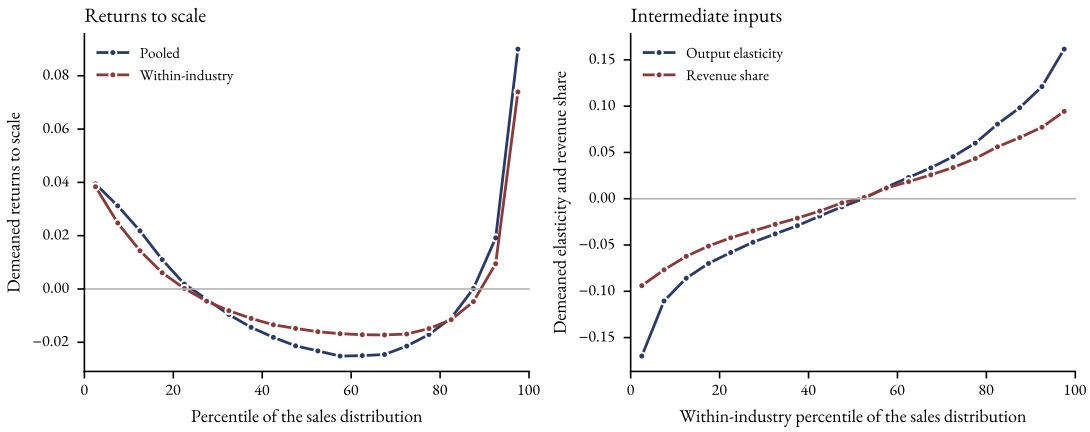
$$\alpha_{ji} = \varepsilon^{m}_{ji} + \varepsilon^{l}_{ji} + \varepsilon^{k}_{ji},$$

where ε^{m}, ε^{l} and ε^{k} are the GNR gross-output materials, labor and capital elasticities, estimated separately for each NAICS2 sector on the Orbis firm-year panel as in Hubmer et al. (2025)[10].

Firm-years are collapsed to one window mean per firm, so F^{α} is a cross section over permanent scalability types, pooled across retained sectors rather than binned by NAICS2. The pooled distribution is winsorized at the 1st and 99th percentiles and discretized to 500 equal-mass nodes. The resulting model-facing support has mean 0.974, standard deviation 0.094, and 32.8% of its mass at $\alpha \geq 1$. In the model, scalability is bounded by the CES markup floor, $\alpha_{ji} < \gamma/(\gamma - 1)$, since cost minimization requires returns to scale not to exceed the markup (Ruzic & Ho, 2023), and above it the anchored equilibrium has no solution. Draws therefore come from F^{α} truncated at $0.99\gamma/(\gamma - 1)$, that is, from the support nodes below this bound with their mass renormalized.[11]

Figure 1 plots mean scalability against firm-year sales percentiles, both pooled across industries within each year and within NAICS2 industry-year cells, in twenty equal-width bins for the 1,391,061 firm-years behind F^{α}. Returns to scale are U-shaped in size: firms in the bottom 5% of their industry's sales distribution have α about 0.038 above the industry mean, the profile falls to about 0.017 below it between the 50th and 75th percentiles, and it then rises steeply to 0.074 above the mean for the top 5%. Ranking firms in the pooled sample gives nearly the same curve, so the profile is not a composition effect across industries. As can be seen in the right panel, the materials output elasticity rises from 0.170 below the industry mean for the smallest firms to 0.162 above it for the largest, while the materials revenue share rises from 0.094 below to 0.096 above. The labor elasticity falls with size and lifts the left arm of the U. This is the shape Hubmer et al. (2025) find for U.S. and Canadian manufacturing, where returns to scale are also U-shaped

Figure 1: Returns to scale and intermediate inputs by sales percentile.



Note. Mean returns to scale (left) and mean output elasticity and revenue share of intermediate inputs (right) by sales percentile for 1,391,061 Orbis firm-years of 2015–2019. The pooled curve ranks firm-years across industries within each year and demeans returns to scale by the full-sample mean. The within-industry curves rank and demean firm-years within NAICS2 industry-year cells. The plotted within-industry RTS gap is 0.071. The 0.046 calibration target instead ranks firms by mean sales within NAICS4 markets and uses mean RTS per firm. F^{α} also uses firm-mean RTS.

[10]The same estimates deliver the value-added weight $\phi_v = 1 - \bar{\varepsilon}^{m}/\bar{\alpha} = 0.517$ of Table B1.

[11]At the calibrated γ the bound is 1.134, which excludes 5.2% of the support mass.

in revenue and the steep increase at the top is also driven by the intermediate input elasticity, albeit with a less pronounced left arm than in the Spanish data.

In the model, α alone does not deliver a size ordering: at the anchor $\hat{y}$ marginal cost is independent of α, and a higher α lowers marginal cost above the anchor but raises it below, so scalability by itself spreads firms around $\hat{y}$ rather than ranking them by size. Their position in the size distribution therefore depends on the alignment between scalability and productivity that $\bar{\rho}$ governs in (9), which I calibrate in Section 3.3 to the gap in mean α between the top 5% and the bottom 50% of firms by sales measured on firm means within NAICS4 markets following Hubmer et al. (2025), which for the Spanish Orbis data is around 0.046.

3.3. Calibration

Parameters and targets. In the (α, ν)-economy baseline the calibrated vector is $(\xi, N, \gamma, \bar{\rho})$, which targets four pooled scalars, namely the aggregate cost-weighted markup $\mu_{\text{cw}}^{\text{data}} \approx 1.18$, the firm-sales tail indices ζ at the top 5% and top 2%, and the returns-to-scale gap between the top 5% and the bottom 50% of firms by sales. The within-market elasticity γ controls the common substitution environment and helps pin the markup level through the markup target, whose model analogue is

$$\mu_{\text{cw}}^{\text{model}} = \int_0^1 \sum_{i \in A_j} \lambda_{ji}^c \mu_{ji} dj, \quad \lambda_{ji}^c \equiv \frac{TC_{ji}}{\int_0^1 \sum_{h \in A_j} TC_{jh} dj}$$

, where λ_{ji}^c is firm i's share of aggregate variable cost. The across-market elasticity η governs cross-sector substitution and is fixed at 1.2, a value within the range of across-market elasticities in the literature (Atkeson & Burstein, 2008; De Loecker et al., 2021; Edmond et al., 2015). The Pareto tail index ξ of the fundamental productivity ν and the Poisson mean N, which sets the mean number of firms in a sector, jointly discipline the upper tail of firm sales through the two tail indices, a choice motivated by the firm-size evidence of Axtell (2001) and Gabaix (2009), and by the Zipf calibration in Di Giovanni & Levchenko (2012). On the data side, these indices are OLS estimates on firm-mean sales over 2015–2019, normalized by the NAICS2 mean of firm means, and the model applies the same estimator to pooled active-firm sales. Orbis concentration measures are avoided as calibration targets, since Kalemli-Özcan et al. (2024) show that measured industry concentration depends on the representativeness of the firm sample, and concentration measures in general tend to be highly dependent on market definition, which is by itself subjective and can lead to severely different results[12].

The rank-copula parameter $\bar{\rho}$ sets the strength of the α–ν dependence and is calibrated to the within-market returns-to-scale gap of 0.046, the mean demeaned α of the top 5% minus the bottom 50% of firms by sales within each NAICS4 market. This follows Hubmer et al. (2025), who use the same top-5%-versus-bottom-50% RTS-gap statistic to help discipline the relationship between TFP and returns to scale in their calibration. They also meet the problem that the joint distribution of returns to scale and TFP cannot be read off the production-function estimates[13], and choose to calibrate the correlation between the two internally rather than imposing the estimated one. The productivity scale $\underline{\nu}$ is not part of the calibrated vector, being solved by the baseline anchoring condition $\tilde{y} = \hat{y} \equiv 1$, which centers the α-gradient at the operating scale rather than fighting the returns-to-scale gap moment. Objects treated as externally disciplined are the pooled scalability distribution F^{α}, the firm-level gross-output returns to scale α_{ji}, the value-added weight ϕ_v, and the common capital share $a = 1/3$ taken from Edmond et al. (2023). See Table B1 for an overview of all parameters and their final values.

[12]See Edmond et al. (2015), De Loecker et al. (2021) among others.

[13]Rescaling output and inputs by a factor s shifts estimated log TFP $\ln \hat{z}_{ji}$ by the firm-specific amount $(1 - \alpha_{ji}) \ln s$, such that relative TFP across firms has no cardinal interpretation.

Table 1: Calibrated parameters and model fit.

Parameter	Value	Target moment	Data	Model	Dev. (%)
γ (within-market elast.)	7.88	μ_{cw}	1.1815	1.1816	0.001
ξ (Pareto tail of ν)	9.50	Sales tail index ζ, top 5%	1.1079	1.1078	0.008
N (Poisson mean firms)	604.0	Sales tail index ζ, top 2%	1.1451	1.1452	0.014
$\bar{\rho}$ (rank copula α–ν)	−0.166	RTS gap, top 5% − bottom 50%	0.0460	0.0460	0.007
η (across-market elast.)	1.2	fixed, not calibrated			

Note. The four calibrated parameters are solved jointly, while η is fixed at 1.2. The row pairing is for presentation: both sales-tail indices jointly discipline ξ and N. Dev. is the absolute deviation of the model moment from its target, in percent of the target. Full parameterization in Appendix Table B1.

Calibration Results. As shown in Table 1, all four moment targets are hit to within 0.015%, with the largest deviation being for the top-2% firm-sales tail index, around 0.014% off. As for the resulting parameter values, the within-market elasticity $\gamma = 7.88$ implies a common CES markup floor of 1.145. Thus, γ supplies most of the average markup wedge through the common CES floor (80.1%), with the remaining 19.9% coming from endogenous share-driven market power.

Moving on to the Pareto tail parameter $\xi = 9.50$, note that it is a tail on productivity $\nu \propto z^{\frac{1}{\alpha}}$ and not on the TFP coefficient z, whose implied distribution is a mixture of Paretos with indices ξ/α_{ji}, such that ξ is not directly comparable with tail parameters on TFP under constant returns to scale.

Matching the two firm-sales tail indices in the Cournot economy yields a Poisson mean of $N = 604$. This remains the mean number of firms competing within a market, but the calibration identifies it from the pooled sales tails rather than an observed firm count per market. Changing N changes how productivity draws translate into market shares and the shape of the upper sales tail, so the top 5% and top 2% indices jointly distinguish its effect from that of ξ. In the calibrated economy, many firms hold small shares and charge markups close to the common CES floor, while a few reach the upper tail. Since all allocated firms produce, N includes the small-firm fringe.[14]

Finally, the rank-copula parameter $\bar{\rho} = -0.166$ corresponds to $\text{corr}(\nu, \alpha) \approx -0.14$ in the calibrated economy. Negative sorting places some high-α firms with low-ν firms at the small end of the sales distribution. Among firms that reach output above the anchor, higher α lowers marginal cost as they expand, allowing high-α firms to appear again at the top. This mechanism is consistent with the U-shaped RTS profile in Figure 1. The calibration directly matches the separately constructed top-versus-bottom RTS gap rather than the full profile. Hubmer et al. (2025) also obtain a negative calibrated correlation between TFP and returns to scale.

4. Welfare Analysis

4.1. Welfare Cost of Markups

Markup Distribution and Welfare Cost. The calibrated $\gamma = 7.88$ implies a common CES floor of $\mu_{\min} = \frac{\gamma}{\gamma-1} \approx 1.145$. Against the fitted aggregate markup of 1.182, this floor contributes 80.1% of the average wedge above one and the share-driven component of (8) contributes 19.9%. Weighting firms by cost, the 25th, 50th and 75th percentiles of firm markups, 1.146, 1.148 and 1.175, lie close to the floor. The calibration matches sales tails from a broad Spanish Orbis sample. As mentioned above, its $N \approx 604$ includes a large fringe of small active firms without a selection margin to drop them, leaving many Cournot shares small and markups near the floor. Firms with larger shares still form a thin upper tail. At the 90th and 99th percentiles, firm markups are 1.266 and 1.529, against 1.221 and 1.331 for sector markups, defined as the cost-weighted

[14] As a descriptive model diagnostic, the sales-weighted mean of 1/ HHI across simulated markets is 51.1, compared with 603.7 active firms per market on average. This statistic is computed from model sales shares and is not a calibration target.

mean markup within each sector and ranked by sector cost. Therefore, an aggregate markup of 1.18 can lower welfare through two rather different objects, namely a common wedge that almost every firm charges and a thin tail of large firms that charge far more, and which of the two dominates the welfare impact depends on how the planner would move production across these firms. Removing markups altogether raises steady-state consumption by $\lambda = 32.3\%$ in the calibrated economy, with the planner producing a gross composite Q that is 72.8% larger than in the market, with 114.1% more capital (bottom row of Table 2, Panel A). Since the distribution of markups comes from the heterogeneity of firms, the first question is how much of this cost is due to firms differing in productivity, in scalability or in both.

Counterfactual Economies. To isolate the role of each dimension of heterogeneity, I compare the calibrated economy with three counterfactual economies, each recalibrated on its own free parameters to the targets it can reach, such that the comparison holds observables fixed rather than parameters. The ν-economy holds RTS common at the harmonic mean of F^α and recalibrates (ξ, N, γ), while the α-economy holds productivity common and the homogeneous economy holds both common, which leaves only γ to recalibrate. The calibration results and moment fit are displayed in Table B2 in the Appendix. All four economies match μ_{cw}, whereas only the baseline also matches the two Zipf indices and the ν-economy nearly matches them, since a bounded RTS distribution generates no Pareto tail in sales.[15] Without a fat tail no firm reaches a sizeable share, such that markups collapse onto the CES floor and γ is pinned by the near-atomistic limit $\gamma/(\gamma-1) \approx \mu_{\text{cw}}$, which is why the homogeneous and α-economies share $\gamma = 6.556$. As for the ν-economy fit, without heterogeneity in α, the Zipf indices must come from ν alone, which requires a fatter productivity tail ($\xi = 5.05$ against 9.50). Within-market shares, and so μ_{cw}, then rise, and the fit offsets this by spreading sales over more firms per market ($N = 1998$).

Which Heterogeneity Matters. The most informative comparison is between the ν-economy and the calibrated baseline, which match the same markup level and nearly the same

Table 2: Productivity, scalability, and the welfare cost of markups.

	Planner vs. market (% change)		
Panel A. Economy	Q	K	C **(welfare** λ**)**
Homogeneous Economy	32.39	56.42	10.77
ν-Economy	36.46	61.28	14.25
α-Economy	39.13	67.76	12.96
(α, ν)-Economy (Calibrated Baseline)	**72.78**	**114.12**	**32.04**
	Contributions (pp; shares in parentheses)		
Panel B. Component	Q	K	C **(welfare** λ**)**
Common level (both homogeneous)	32.39 (44.5%)	56.42 (49.4%)	10.77 (33.6%)
Productivity dispersion, α held common	4.07 (5.6%)	4.86 (4.3%)	3.48 (10.9%)
RTS dispersion, ν held common	6.74 (9.3%)	11.34 (9.9%)	2.19 (6.8%)
Joint heterogeneity interaction	**29.58 (40.6%)**	**41.50 (36.4%)**	**15.60 (48.7%)**
Total (Calibrated Baseline)	**72.78 (100%)**	**114.12 (100%)**	**32.04 (100%)**

Note. Panel A compares each economy's planner with its own market, where Q is the gross composite (2), K is capital and the change in consumption C equals the consumption-equivalent welfare gain $\lambda = \frac{C^{PE}}{C^{ME}} - 1$ by (26), all in percent. Panel B splits the calibrated gains into percentage-point contributions (shares in parentheses). Each counterfactual economy is recalibrated to match μ_{cw}, with the ν-economy (common scalability at the harmonic mean of F^α) refitting (ξ, N, γ) and also nearly matching the Zipf indices of sales, while the α- and homogeneous economies (common productivity) refit only γ, since a bounded scalability distribution generates no Pareto tail. All economies use $M = 5000$ markets and their own χ. The joint interaction is distinct from the sorting effect in Table 3.

[15] With common ν, size differences come only from α, and the capped, data-disciplined F^α keeps every firm away from the singularity at $\gamma/(\gamma-1)$ where size diverges, so sales have no Pareto tail. Hubmer et al. (2025) obtain one only by letting returns to scale approach that bound.

sales tails with almost the same γ (7.78 against 7.88). Adding the observed heterogeneity in scalability on top of the ν-economy more than doubles the welfare cost of markups, from 14.3% to 32.3%, while the homogeneous economy, at 10.8%, gives the cost of a common markup of 1.18 alone (Table 2, Panel A). The output and capital responses of the planner follow a different ranking, as the α-economy expands the gross output by 39.1% and capital by 67.8%, slightly more than the ν-economy (36.5% and 61.3%) despite its smaller welfare gap, while the baseline expands both by about twice as much as the homogeneous economy (72.8% and 114.1%, against 32.4% and 56.4%). In every economy capital grows faster than output and output faster than consumption (equivalent to λ as per (26)), since the larger capital stock and materials use of the planner are paid for out of Q, with $C = Q - M - \delta_K K$ in steady state, which suggests that heterogeneity in RTS raises the welfare cost mostly through how much more the planner would produce, a point to which I return in Section 4.2.

Moving on to Panel B, the calibrated gain is split into the gain of the homogeneous economy, the gains of the ν- and α-economies over it and a residual joint interaction, such that the four contributions sum to λ by construction[16]. Each individual heterogeneity margin contributes little on its own, with 3.48 percentage points from productivity dispersion and 2.19 from scalability dispersion. As in Panel A, RTS heterogeneity has the biggest contribution to the capital and therefore the Q gap. Likewise, productivity dispersion ends up having the biggest welfare role out of the two again, potentially due to the lack of an expressive Pareto tail under RTS dispersion only. The joint interaction of the two accounts for 15.60 percentage points, or 48.7% of the calibrated gain, such that almost half of the welfare cost of markups under the calibrated baseline arises only when firms differ along both dimensions at once. Understanding what is behind this interaction effect requires splitting a welfare gain into the channels through which wedges lower consumption, which I turn to next.

4.2. Decomposing the Welfare Cost

Welfare Setup. To investigate the components behind the interaction effect, I define and work with the log of the gain λ of (26),

$$\Delta \equiv \ln(1+\lambda) = \ln C^{PE} - \ln C^{ME}, \tag{28}$$

which is additive across any chain of intermediate allocations between the market and the planner, such that each decomposition below is an exact identity rather than a local approximation. Every intermediate allocation is a steady state solved on the same draws (α_{ji}, ν_{ji}), the same active firm sets and the same frozen anchor $\hat{y}$ as the market and the planner. Additionally, it shares their normalizations $P = 1$ and $L = 1$ and the market value of χ, such that allocations differ only in pricing and in the constraints they place on aggregate inputs.

The first decomposition asks whether markups lower welfare through their common level or through their dispersion across firms, as is common in the literature (Baqaee & Farhi, 2020; De Loecker et al., 2020; Edmond et al., 2023). To decompose the total effect into these two components, I introduce the uniform-markup allocation (U), where every firm charges the common cost-weighted average markup $\bar{\mu} = \mu_{\text{cw}}^{\text{model}}$ of the market (34), such that the aggregate wedge is preserved by construction and only its dispersion across firms is removed. The decomposition then splits the gain into a *dispersion* component, from the market to U, and a *level* component, from U to the planner, such that

$$\Delta = \Delta_{\text{disp}} + \Delta_{\text{level}}, \quad \Delta_{\text{disp}} = \ln\left(\frac{C^U}{C^{ME}}\right), \quad \Delta_{\text{level}} = \ln\left(\frac{C^{PE}}{C^U}\right). \tag{29}$$

[16]The contributions are λ^{hom}, $\lambda^{\nu} - \lambda^{\text{hom}}$, $\lambda^{\alpha} - \lambda^{\text{hom}}$ and $\lambda^{\alpha\nu} - \lambda^{\nu} - \lambda^{\alpha} + \lambda^{\text{hom}}$. Since every economy is recalibrated, each contribution compares economies that match the same targets, rather than switching one dimension off at given parameters.

The second decomposition asks instead whether the planner gains by reallocating the inputs (i.e. fixing misallocation) that the market already uses or by using more of them, in the spirit of Hsieh & Klenow (2009). The dispersion of revenue productivity, which Hsieh & Klenow (2009) use to measure distortions, no longer maps into misallocation once returns to scale differ, since undistorted firms then differ in revenue productivity as well (Ruzic & Ho, 2023). I therefore compute the misallocation component from the counterfactual allocation itself. To explore these different components, I introduce two constrained planners, both pricing at marginal cost. The fixed-input planner (PE_I) holds aggregate capital and materials at their market levels, $K = K^{ME}$ and $M = M^{ME}$. Therefore, the rental rate R rises above the Euler rate (1) and a shadow materials price P_{mat} takes the place of $P = 1$ inside the unit cost (14), each adjusting until aggregate demand for its input returns to the market level. Since both prices enter the unit cost of every firm, this re-prices the whole cross section rather than relabeling the planner allocation. Finally, the fixed-capital planner (PE_K) frees materials at $P = 1$ while still holding capital at K^{ME}, and the input decomposition splits the same gain into *misallocation*, *materials* and *capital* components, such that

$$\Delta = \Delta_{\text{misalloc}} + \Delta_{\text{mat}} + \Delta_{\text{cap}}, \tag{30}$$

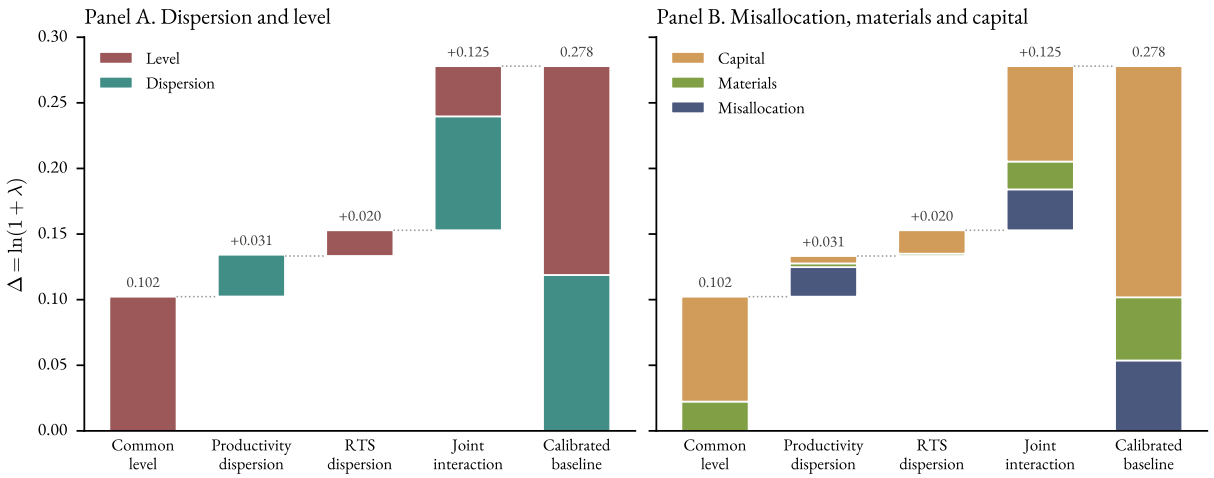
$$\Delta_{\text{misalloc}} = \ln\left(\frac{C^{PE_I}}{C^{ME}}\right), \quad \Delta_{\text{mat}} = \ln\left(\frac{C^{PE_K}}{C^{PE_I}}\right), \quad \Delta_{\text{cap}} = \ln\left(\frac{C^{PE}}{C^{PE_K}}\right).$$

Note that the two decompositions are not nested, as the first cuts the gain by which wedges are removed and the second by which inputs are allowed to move. Both, however, sum exactly to the same Δ, such that the share of each component in Δ is well defined and the shares add up to one within each decomposition.

From the Homogeneous Economy to the Baseline. In Figure 2, the welfare gain of each economy is stacked by both decompositions, moving from the homogeneous economy to the calibrated baseline in the steps of Panel B of Table 2. The homogeneous economy has $\Delta = 0.102$, all of it level, since its markups are uniform, and mostly capital (0.080), with the remaining 0.022 coming from materials. Productivity dispersion then adds 0.031, essentially all of it through dispersion (0.032) in Panel A and mostly through misallocation (0.022) in Panel B, whereas RTS dispersion adds 0.020, all of it through the level and mostly through capital (0.018), with no dispersion component at all, since the markups of the α-economy are near uniform as well, as discussed above. Finally, the interaction contributes nearly half (0.125) of the final calibrated $\Delta = 0.278$, and 70% of it comes through dispersion (0.087) in Panel A, while capital takes 58% of it (0.073) under the input decomposition in Panel B.

The calibrated bar then shows the composition of the full gain. The dispersion component accounts for 0.119 of Δ, or 43%, with the remaining 0.159 coming from the common level, while the input decomposition attributes 0.054 to misallocation, 0.048 to materials and 0.176, or 63%, to capital. That is, reallocating the inputs that the market already uses would raise consumption by 5.5%. This is the counterpart of the fixed-input gains of Hsieh & Klenow (2009), who find manufacturing TFP gains of 30–43% from equalizing revenue productivity in the United States, but it is much smaller, since only the dispersion created by markups is removed here, while their measure captures every source of dispersion in revenue productivity. Most of the gain, however, comes from the planner accumulating capital at the Euler rate, which amplifies fixed-input gains as in their exercise, and does so far more here, since the planner also removes the common markup level that equalizing revenue productivity leaves in place. Restuccia & Rogerson (2008) draw a similar distinction, holding aggregate capital fixed to isolate the reallocation effect of distortions, and find that letting capital adjust adds a further output loss through lower accumulation. Since all comparisons are across steady states, the capital component ignores the consumption forgone

Figure 2: From the homogeneous economy to the calibrated baseline, by channel of the welfare gain.



Note. Bars walk along the welfare gain $\Delta = \ln(1+\lambda)$ from the homogeneous economy to the calibrated baseline in the steps of Panel B of Table 2, namely common level (Δ^{hom}), productivity dispersion ($\Delta^{\nu} - \Delta^{\text{hom}}$), RTS dispersion ($\Delta^{\alpha} - \Delta^{\text{hom}}$) and the joint interaction ($\Delta^{\alpha\nu} - \Delta^{\nu} - \Delta^{\alpha} + \Delta^{\text{hom}}$). Each step is stacked by the components of one decomposition, which sum to Δ exactly in every economy, namely the markup decomposition (29) into dispersion ($ME \to U$) and level ($U \to PE$) in Panel A, and the input decomposition (30) into misallocation ($ME \to PE_I$), materials ($PE_I \to PE_K$) and capital ($PE_K \to PE$) in Panel B.

along the transition to a capital stock more than twice as large, and λ should be read as an upper bound on the gain net of accumulation.

Complementarity. The two dimensions therefore act as complements. Productivity generates the fat tail of sales, and with it the large shares and high markups, while scalability shapes both how far a firm grows and how much output a given markup withholds from it, such that the cost of markups is largest where high productivity and high RTS meet. The markups of the ν-economy and the baseline make this point sharper. The cost-weighted variance of log markups is higher in the ν-economy than in the calibrated baseline, namely 0.0042 against 0.0037, and yet its dispersion component is about a quarter of the baseline (0.032 against 0.119), such that the cost of dispersion depends on how scalable the firms carrying them are, and not only on how dispersed firm markups are.

A potential explanation comes from the marginal cost (16). Holding the sector aggregates fixed and abstracting from the feedback of shares on markups, a firm facing the within-market elasticity γ responds to a change in its markup with

$$\varepsilon_{\mu} \equiv \frac{d \ln y_{ji}}{d \ln \mu_{ji}} = -\frac{\gamma \alpha_{ji}}{\alpha_{ji} + \gamma(1 - \alpha_{ji})}, \tag{31}$$

which rises with α_{ji}, as the price elasticity of supply increases with returns to scale (Ruzic & Ho, 2023), and diverges as α_{ji} approaches $\gamma/(\gamma-1)$, the bound below which the scalability draws are capped. Under constant returns, the elasticity ε_{μ} defined in (31) equals the calibrated $\gamma = 7.88$ but rises to about $\varepsilon_{\mu} = 28$ at $\alpha_{ji} = 1.1$, a value well inside the support of F^{α}. A given wedge thus withholds far more output when it sits on a highly scalable firm, and the RTS gap of 0.046 that the calibration matches places such firms at the top of the sales distribution, where Cournot markups are highest. The convexity of (31) in α_{ji} could also explain why scalability dispersion alone raises the level component, since the average response to removing a common wedge grows with the spread of α. If the cost of markups is largest where high productivity meets high RTS, it should also depend on how the two are paired across firms, which the calibration disciplines through the RTS gap. The next natural question is then to ask how impactful the alignment/misalignment between productivity and RTS is for the welfare cost of wedges.

4.3. Sorting and the Cost of Markups

Sweep. The interaction in Table 2 shows that the welfare cost of markups depends on productivity and scalability jointly, but it does not show how much it depends on the way the two are paired across firms. To answer this, I hold the calibrated parameters and both marginal distributions fixed and vary only the strength $\bar{\rho}$ of the rank copula (9). The experiment re-solves the economy for values of $\bar{\rho}$ between -0.9 and $+0.9$, keeping (ξ, N, γ, η), the scalability draws, the anchor $\hat{y}$ and the baseline χ at their calibrated values, such that the productivity marginal is preserved in distribution. Additionally, three arrangements permute the calibrated draws within each sector, namely perfect negative and positive rank pairings, labelled $\bar{\rho} = \mp 1$ in Table 3, and a random reassignment that removes any dependence, all of which preserve both marginals exactly. Two features set this experiment apart from Table 2. First, the arrangements are not recalibrated, such that they no longer match the RTS gap, the Zipf indices or μ_{cw}. The sweep therefore measures how the welfare cost responds to the pairing at given primitives, while Table 2 compares economies that agree on the observed moments. Second, the markup decomposition re-solves U at each arrangement's own μ_{cw}, such that the level component absorbs any change in aggregate market power and the dispersion component captures the cost of markups differing across firms at a given level.

How Much the Pairing Matters. The pairing of scalability and productivity moves the welfare cost of markups by more than any dimension of heterogeneity in Table 2. As shown in Table 3, λ rises from 9.0% under perfect negative pairing to 152.3% under perfect positive pairing[17], with random reassignment at 50.3%. This echoes Restuccia & Rogerson (2008), who show that idiosyncratic distortions are far more costly when they are correlated with establishment productivity than when they are not, although here the correlation between markups and scalability arises in equilibrium rather than being imposed. The mildly negative sorting of the calibrated economy (-0.14) keeps the cost well below that of independent assignment or even positive pairing. The response is clearly S-shaped, as can be visually confirmed in Panel A of Figure B1 in the Appendix, being flat under strongly negative sorting, steepest around the calibrated point and flattening again at the top, where the last copula point ($\text{corr}(\alpha, \nu) = 0.71$) lies marginally below the previous one and only the exact permutation reaches 152.3%. Around the calibrated economy, λ doubles between a correlation of -0.35 and -0.14 (from 16.0% to 32.3%)

Table 3: Sorting and the welfare cost of markups.

		Market markups			Planner vs. market (% change)		
Arrangement $\bar{\rho}$	$\text{corr}(\alpha, \nu)$	μ_{cw}	$\text{Var}(\ln \mu)$	$\text{corr}(\ln \mu, \alpha)$	Q	K	**Welfare**
Lowest: $\bar{\rho} = -1$	-0.92	1.147	0.00	–	28.7	49.9	9.0
Sweep: $\bar{\rho} = -0.60$	-0.54	1.149	0.01	0.20	31.6	54.0	10.3
Sweep: $\bar{\rho} = -0.40$	-0.35	1.157	0.08	0.37	43.2	71.3	16.0
Calibrated $\bar{\rho} = -0.166$	**-0.14**	**1.182**	**0.37**	**0.48**	**73.3**	**114.9**	**32.3**
Random reassignment	**≈ 0.00**	**1.210**	**0.72**	**0.51**	**104.7**	**160.1**	**50.3**
Sweep: $\bar{\rho} = +0.30$	$+0.25$	1.271	1.41	0.50	175.8	264.2	91.3
Sweep: $\bar{\rho} = +0.60$	$+0.48$	1.324	1.76	0.46	240.3	361.2	127.6
Sweep: $\bar{\rho} = +0.75$	$+0.60$	1.339	1.69	0.43	258.6	388.8	137.6
Highest: $\bar{\rho} = +1$	$+0.77$	1.356	1.54	0.34	284.9	427.4	152.3

Note. Q is the gross composite (2) and K capital, with materials expanding by the same percentage as capital. Changes compare planner and market steady states within each row at fixed labor and firm counts, such that the welfare column equals the consumption increase. Rows are indexed by the copula parameter $\bar{\rho}$, next to the realized $\text{corr}(\alpha, \nu)$, with $\bar{\rho} = \pm 1$ the perfect rank pairings obtained by within-sector permutation and random reassignment the average of two permutations. All rows keep the baseline anchor and χ and use $M = 10000$ markets, against $M = 5000$ in Table 2, which is why the calibrated row differs slightly from that table. Markups are over active firms and cost-weighted, with $\text{Var}(\ln \mu)$ multiplied by 100, and their correlation with α is undefined at $\bar{\rho} = -1$, where all markups sit at the CES floor.

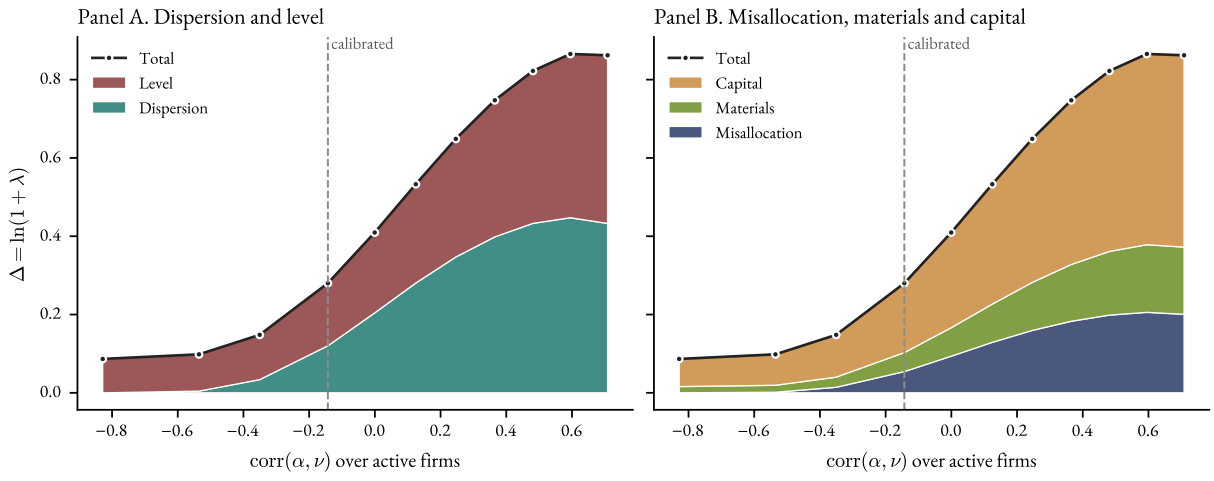
[17]The perfect pairings, labelled $\bar{\rho} = \mp 1$, are boundary cases rather than plausible economies, and they serve to bound the welfare cost of markups at the calibrated shared marginals.

and rises by a further 18 percentage points to random assignment, which makes $\bar{\rho}$, and the RTS gap that identifies it, a first-order determinant of the welfare cost under RTS and productivity heterogeneity. Note that concluding that sorting decreases welfare would be incorrect. In fact, Panel B of Figure B1 shows that sorting raises welfare in both allocations, as lifetime welfare of the market and of the planner both rise with $\text{corr}(\alpha, \nu)$. However, the planner rises faster, which widens the gap, and thus the welfare gain from going from market to planner. Pairing scalable with productive firms thus raises what the economy can produce, while markups withhold a growing share of it.

Sorting also creates market power. Since aligning scalability with productivity lets the most productive firms expand further, their shares grow and μ_{cw} rises from 1.147 to 1.356 along the sweep, with the CES floor of 1.145 unchanged. Markups also differ more across firms, as the variance of log markups in Table 3 rises almost fivefold from the calibrated point to $\bar{\rho} = +0.6$. Their correlation with scalability, by contrast, rises only up to random assignment, from 0.20 at $\bar{\rho} = -0.6$ to about 0.5, and falls to 0.34 under perfect positive pairing. Part of the rise in λ must therefore come from a higher markup level, and part from markups differing more across firms. At the same time, the capital that the planner would add grows from 49.9% to 427.4% of the market stock along the sweep, with materials likewise expanding (Table 3). This indicates that the cost may lie less in how the market allocates the inputs it uses than in the inputs it withholds altogether. The two decomposition strategies presented in Section 4.2 are applied here to quantify these different channels.

Decomposing Sorting. I reuse the welfare decomposition machinery from Section 4.2, solving the intermediate planner allocations (U, PE_I, PE_K) at each arrangement. Since U is re-solved at each arrangement's own μ_{cw}, the level component of the markup decomposition in Panel A of Figure 3 absorbs the higher markup level, while the dispersion component captures their spread across firms, and the latter rises from zero under strongly negative sorting to 0.12 at the calibrated point and to about 0.45 at the top, around half of the total Δ. Moving on to Panel B, the misallocation component rises from zero to about 0.2, or roughly a quarter of Δ, since a common markup leaves nothing to reallocate at fixed inputs, while capital remains the largest component throughout, such that sorting raises the cost of markups both by misplacing the inputs already in use and, above all, by holding back the expansion of scalable firms. Up to random assignment, both rises go along with a wider spread of markups and a growing correlation

Figure 3: Decomposition of the welfare gain from the planner along the copula sweep.



Note. Each point is a solved economy plotted at its realized $\text{corr}(\alpha, \nu)$, as in Figure B1, and the dashed lines mark the calibrated economy. The black line is the total steady-state gain $\Delta = \ln(1 + \lambda)$, and the shaded components sum to it exactly. Panel A splits it by (29) into dispersion $(ME \to U)$ and level $(U \to PE)$, with U re-solved at each economy's own μ_{cw}. Panel B splits it by (30) into misallocation $(ME \to PE_I)$, materials $(PE_I \to PE_K)$ and capital $(PE_K \to PE)$. Labor, firm sets, the anchor and χ are fixed throughout, and areas between points are interpolated.

between markups and scalability in Table 3. Beyond it, the correlation falls while the spread keeps widening up to $\bar{\rho} = +0.6$. Past that point the spread narrows slightly, yet the dispersion component does not decline, since sorting keeps moving inputs towards scalable firms, whose output responds most to a given markup by (31). The cost-weighted mean of $|\varepsilon_\mu|$ rises from 17 at the calibrated point to 70 under perfect positive pairing, such that the same markups withhold more output.

5. Conclusion

The modern literature on market power usually explains markups by productivity under common returns to scale. I introduce heterogeneity in RTS, measured from Orbis Spain using the Gandhi et al. (2020) estimation method, to an Atkeson & Burstein (2008)-style Cournot oligopoly framework, and link it to productivity through a calibrated rank copula. Once firms differ in RTS, the same markup withholds very different amounts of output, so shares and markups alone cannot measure the cost of market power accurately.

Under the baseline calibrated (α, ν)-economy, I find that the cost of markups λ equals 32 percent, more than double that of the ν-economy, which matches the same markup level and nearly the same sales tails with common returns to scale. Each dimension of heterogeneity contributes little on its own, with the joint interaction of the two accounting for almost half of the gap, such that productivity and scalability act as complements. The reason is that the two margins play different roles, as productivity drives the large shares and thus the high markups, while scalability determines how much output a given markup withholds, such that markups are most costly at firms that are both productive and scalable.

To investigate the components of this welfare gap, I split it in two separate ways. The first separates the dispersion of markups across firms, which accounts for a substantial share of the gap, from their common aggregate level, which accounts for the larger remainder. The second separates the misallocation of the inputs that the market already uses, which plays only a minor role, from the underuse of materials and capital, which accounts for most of the gap, with capital alone being its largest component. That is, markups in this framework lower welfare mostly by holding back the expansion of scalable firms.

Moving on to sorting, varying only the pairing of the two margins, while holding both marginal distributions fixed, moves λ across a range far wider than that spanned by any dimension of heterogeneity in Table 2, with the calibrated economy in the lower part of it. Sorting also creates market power, as aligning scalability with productivity raises the aggregate markup. Under perfect negative pairing, markups sit close to the CES floor, such that the whole welfare gap comes from their common level in the first split and from the underuse of materials and capital in the second. As the pairing turns positive, the dispersion component grows to about half of the total and misallocation to about a quarter, while capital remains the largest component throughout. Past random assignment, however, the correlation between markups and scalability falls, such that the welfare gap keeps rising due to a wider spread of markups and the reallocation of inputs toward scalable firms, whose output responds most to a given markup.

Some potential caveats qualify these conclusions, however, the first being that the comparison is between steady states, with labor and the number of firms held fixed, such that the transitional costs of reaching the planner's allocation are excluded. Second, scalability and its pairing with productivity are taken as given, while in practice they likely stem from investment in standardization or intangibles (Argente et al., 2025; De Ridder, 2024; Hubmer et al., 2025), and endogenizing them would be the most natural extension. Finally, the evidence comes from Spain alone, and the results remain sensitive to the choice of output anchor. Nonetheless, the results suggest that the welfare cost of markups depends on the expansion technology of the firms that charge them, such that the same markups are far more costly in an economy where the most productive firms are also the most scalable.

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A. Derivations

A.1. Demand, Markup and Cost Derivations

Conditional on market-level expenditure, the CES block in (5) implies the demand system in (6) and the market-share expression in (7). Under Cournot competition, firm i internalizes the effect of its own quantity on the market price index and hence on its residual demand elasticity. The perceived elasticity can be written as

$$\varepsilon_{ji,t}^{-1} = \frac{1 - s_{ji,t}}{\gamma} + \frac{s_{ji,t}}{\eta}, \tag{32}$$

so that

$$\mu_{ji,t} = \frac{\varepsilon_{ji,t}}{\varepsilon_{ji,t} - 1}. \tag{33}$$

Substituting (32) into (33) yields the inverse-markup expression (8) reported in the text. This inverse-markup system is the object that must be solved before aggregation and moment construction.

As a pooled diagnostic, cost-weighting firm revenue-cost wedges across all producing firms yields the aggregate cost-weighted markup

$$\mu_{\text{cw}}^{\text{model}} = \int_0^1 \sum_{i \in A_j} \lambda_{ji}^c \mu_{ji} dj, \tag{34}$$

where λ_{ji}^c is firm i's share of aggregate variable cost. This pooled cost-weighted markup is the model analogue of the aggregate markup target $\mu_{\text{cw}}^{\text{data}}$, which is itself constructed as a cost-weighted mean of firm-level price-marginal-cost ratios. It should not be confused with the accounting ratio of aggregate revenue to aggregate variable cost: because (15) and (18) imply $TC_{ji} = \alpha_{ji} p_{ji} y_{ji} / \mu_{ji}$, that ratio is instead

$$\mu_{\text{cw}}^{\alpha} = \int_0^1 \sum_{i \in A_j} \lambda_{ji}^c \frac{\mu_{ji}}{\alpha_{ji}} dj,$$

which exceeds $\mu_{\text{cw}}^{\text{model}}$ whenever $\alpha_{ji} < 1$, since it also loads on the variable profit that decreasing returns generate even at a unit markup. The two coincide only under constant returns; at the calibrated point they are 1.182 and 1.402. Only $\mu_{\text{cw}}^{\text{model}}$ is targeted.

The cost block follows from two nested Cobb-Douglas cost minimizations, first over capital and labor inside value added, which yields the input ratio in (17) and the unit cost (13), and then over value added and materials, which yields (14) and, through (11), the cost functions (15) and (16).

A.2. Planner Derivations

Let the planner's Lagrangian attach multipliers λ_t^L and λ_t^K to the aggregate labor and capital constraints and let $\psi_{ji,t}$ denote the marginal contribution of one more unit of firm output to the gross composite bundle. The planner's firm-level FOCs are the conditions reported in (23), (24), and (25). Together with (11), they imply

$$U_{C(C_t, L_t)} \psi_{ji,t} \alpha_{ji} \phi_v (1 - a) \frac{y_{ji,t}}{l_{ji,t}} = \lambda_t^L,$$

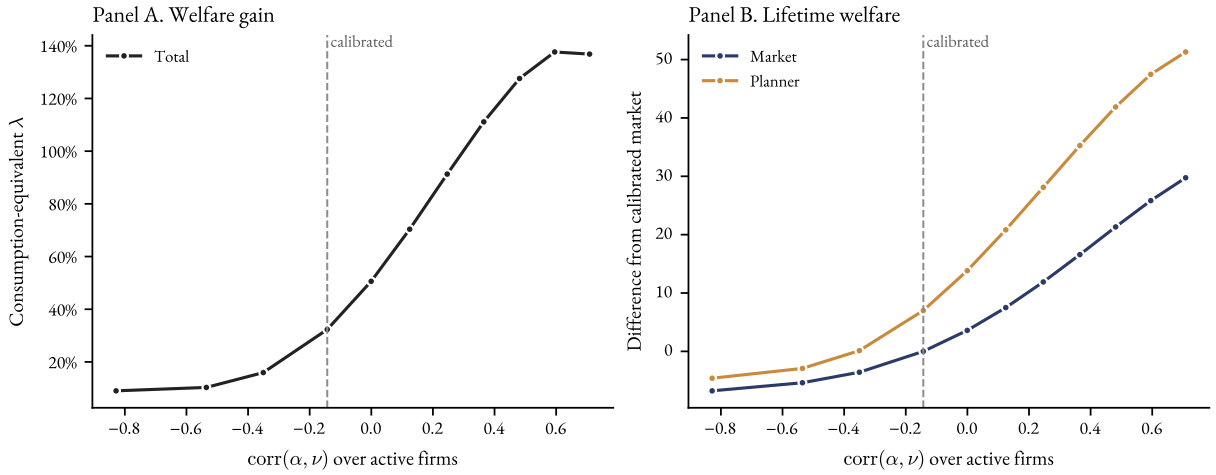
$$U_{C(C_t, L_t)} \psi_{ji,t} \alpha_{ji} \phi_v a \frac{y_{ji,t}}{k_{ji,t}} = \lambda_t^K,$$

$$\psi_{ji,t}(1-\phi_v)\alpha_{ji}\frac{y_{ji,t}}{m_{ji,t}}=1.$$

The planner keeps the same exogenous sector firm set A_j as the decentralized allocation. Hence the planner derivation needs no additional firm-selection condition: all differences between the market and planner allocations come from markup pricing and the induced allocation of labor, capital, materials, and output.

B. Additional Figures and Tables

Figure B1: Consumption-equivalent gains and lifetime welfare against $\mathrm{corr}(\alpha,\nu)$ along the copula sweep.



Note. Each point is a solved economy at a copula strength $\bar{\rho}$, plotted at its realized $\mathrm{corr}(\alpha,\nu)$ over active firms, and the dashed lines mark the calibrated economy. Panel A reports the total steady-state consumption-equivalent welfare gain λ from the market to the planner allocation, as in Table 3. Panel B reports steady-state lifetime welfare as defined in Section 2.5, relative to the calibrated market, namely $W^{r(\bar{\rho})}-W^{ME}(\bar{\rho}_{\mathrm{cal}})$ for $r\in\{ME,PE\}$. With labor and χ fixed, this equals $\frac{\ln\left[\frac{C^{r}(\bar{\rho})}{C^{ME}(\bar{\rho}_{\mathrm{cal}})}\right]}{1-\beta}$. Transitional dynamics are excluded. Both panels use every copula sweep point except $\bar{\rho}=-0.15$, whose realized $\mathrm{corr}(\alpha,\nu)=-0.13$ is indistinguishable from that of the calibrated economy. Segments between points are interpolations.

Table B1: Parameterization used in the pooled sector-market steady-state implementation.

Parameter	Description	Value / Status
A. Assigned parameters		
β	discount factor	0.96
δ_K	capital depreciation rate	0.06
φ	Frisch elasticity of labor supply	1.00
a	common capital share inside value added	0.3333, set to one third
η	across-market elasticity (common)	1.2, fixed from the literature
$\hat{y}$	common anchor output (units normalization)	$\equiv 1$
B. External empirical inputs		
F^{α}	pooled scalability distribution	firm-level gross-output returns to scale
ϕ_v	value-added weight in gross output	0.517, estimated from the GNR elasticities
ω_j	sector exposure weights	normalized to uniform in the current baseline
C. Parameters determined inside the model		
ξ	Pareto tail index of productivity ν	9.50, targets sales tail indices ζ (top 5% and top 2%)
N	Poisson mean firms per sector	604.0, targets sales tail indices ζ (top 5% and top 2%)
γ	within-market elasticity (common)	7.88, targets cost-weighted markup $\mu_{\mathrm{cw}}^{\mathrm{data}}$
$\bar{\rho}$	rank-copula strength of α–ν	-0.166, targets the top-versus-bottom RTS gap
$\underline{\nu}$	productivity scale / Pareto lower bound	8844.41, solved by $\tilde{y}=\hat{y}$, not moment-targeted

Note. The four calibrated parameters $(\xi,N,\gamma,\bar{\rho})$ are solved jointly, with (ξ,N) disciplining the sales tail indices, γ pinning the markup level and $\bar{\rho}$ targeting the returns-to-scale gap. The productivity scale $\underline{\nu}$ is solved by the baseline anchoring condition $\tilde{y}=\hat{y}\equiv 1$ rather than calibrated. External empirical inputs include F^{α}, ϕ_v and ω_j.

Table B2: Calibration of the four economies of Table 2.

	Homogeneous	ν-Economy	α-Economy	(α, ν)-Economy
Parameters				
γ (within-market elast.)	6.556	7.776	6.556	7.875
ξ (Pareto tail of ν)	∞ (fixed)	5.05	∞ (fixed)	9.50
N (Poisson mean firms)	604.0 (fixed)	1997.9	604.0 (fixed)	604.0
$\bar{\rho}$ (rank copula α–ν)	none	none	none	-0.166
α (scalability)	0.957 (common)	0.954 (common)	F^{α}	F^{α}
η (across-market elast.)	1.2 (fixed)	1.2 (fixed)	1.2 (fixed)	1.2 (fixed)
Moments (data)				
μ_{cw} (1.1815)	1.1815	1.1817	1.1815	1.1816
Sales tail ζ, top 5% (1.1079)	55.03*	1.0962	11.96*	1.1078
Sales tail ζ, top 2% (1.1451)	54.55*	1.1584	9.58*	1.1452
RTS gap, top 5% − bottom 50% (0.0460)	0*	0*	-0.030*	0.0460

Note. Each economy is calibrated on its free parameters to the targets it can reach; * marks moments it does not target. A common ν is $\xi = \infty$, and a common α is the harmonic mean of F^{α} below the cap $0.99\gamma/(\gamma - 1)$. With common ν the economy generates no Pareto tail in sales, so no target identifies N, which stays at its calibrated value; $\bar{\rho}$ is undefined once either dimension is common, and η is fixed at 1.2 as in Table 1. N in the ν-economy ends just below its upper bound of 2000.